# CH1 Axioms of Probability

## 1.2 Sample space and events

### Definition

**Experiment**: 一個機率問題是由一個機率事件 (e.g. tossing a die)

**Outcome**: 產生結果 (A.K.A sample point )

**Sample space**: 所有可能的結果集合 (e.g. 丟硬幣 {Head, Tail} )

**Event**: 所有可性的子集合 (e.g. 硬幣頭在上的可能性 {Head} )

### Ex 1.3 (An infinite sample space)

Measuring the lifetime of a light bulb, where last at least 100 hrs

$S = \{x : x \geq 0\} \quad E = \{x : x \geq 100\}$

### Ex 1.4 (An finite sample space) (a.k.a **Classical Probability**)

All families with 1, 2, or 3 children (genders specified)

$S = \{b, g, bg, gb, bb, gg, bbb, bgb, bbg, bgg, ggg, gbg, ggb, gbb\}$

### Definition

Occur: Event E has occured if result contains a element in E

| | |
|---|---|
| $E \subset F$ | E is a subset of F |
| $E \cap F = EF$ | Intersection of E and F |
| $E \cup F$ | Union of E and F |
| $E - F$ | The set difference of E and F |
| $E^c = \bar{E}$ | All element excluding E = $S - E$ |
| $\cup_{i=1}^{n} E_i$ | $E_1 \cup E_2 \cup E_3 ... E_n$ |
| $\cap_{i=0}^{n} E_i$ | $E_1 \cap E_2 \cap E_3 \cap ... \cap E_n$ |

$,\cup_{i=1}^{\infty} E_i, \cap_{i=1}^{\infty} E_i$

### Sample space and events

Commutative laws (交換律) EUF=FUE , EF=FE

Associative laws (結合律) EU(FUG)=(EUF)UG

**Distributive laws** (分配律) $(E \cup F)H = (EH) \cup (FH)$

$(EF)U \cup H = (E \cup H)(F \cup H)$

De Morgan's law 1st rule $(E \cup F)^c = E^c F^c$

2nd rule $(EF)^c = E^c \cup F^c$

E and F are **mutually exclusive** if they have nothing in common $EF = \phi$

無中生有 $E = ES = E(F \cup F^c) = EF \cup EF^c$ (也是因為 $EF$與 $EF$是mutually exclusive)

## 1.3 Axioms of probability

S (sample space): 所有的可能性的集合

A (event): 其中可能性的集合

Pr (Function): 發生 A 事件的機率為 P(A)

Axioms 1: P(A) ≥ 0 機率必定大於 0

Axioms 2: P(S) ≤ 1 機率最大為 1

Axioms 3: 只有在所有 event 皆 mutually exclusive 時，機率才為每個 event 的和

If $\{A_1, A_2, A_3, ..., A_n\}$ are mutually exclusive then: $P(\cup_{i=0}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i)$

Classical definition of Probability

1. Sample space is finite
2. Each event is equally likely to occur (shares a same probability)

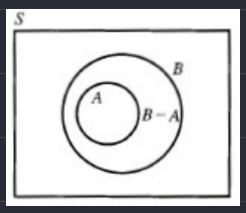
$P(A) = N(A)/N(S)$ ... N(A) is the number of sample points of A.

## 1.4 Basic Theorem

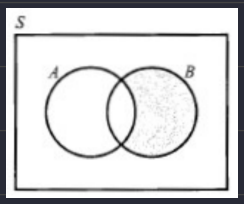
**Theorem 1.4:** $P(A^c) = 1 - P(A)$

**Theorem 1.5: If** $A \subset B$**, then** $P(B - A) = P(BA^c) = P(B) - P(A)$

**Corollary If** $A \subset B$**, then P(A) ≤ P(B)**




**Theorem 1.6:** $P(A \cup B) = P(A) + P(B) - P(AB)$




- **Inclusion-Exclusion Principle**

$$P(A_1 \cup A_2 \cup \ldots \cup A_n) = \sum P(A_i) - \sum P(A_i A_j) + \sum P(A_i A_j A_k) - \ldots + (-1)^{n-1} P(A_1 A_2 \ldots A_n)$$

1.5 Continuity of probability function (Omission)

- **Recall the continuity of a function f: R→R**

$$\lim_{n\to\infty} f(x_n) = f(\lim_{n\to\infty} x_n)$$

**for every convergent seq $\{x_n\}$ in R.**

- **The continuity of probability function is similar.**
- **Def. A seq $\{E_n, n>=1\}$ of event of a sample space is called increasing if**

$$E_1 \subseteq E_2 \subseteq E_3 \subseteq \cdots E_n \subseteq E_{n+1} \subseteq \cdots;$$

**it is called decreasing if**

$$E_1 \supseteq E_2 \supseteq E_3 \supseteq \cdots E_n \supseteq E_{n+1} \supseteq \cdots.$$

## 1.6 Probabilities 0 and 1